

Course Code: CAP 456	Course Instructor: Dr. Apash Roy	
Course Title: Big Data	Roll Number: RD2110B61	
Name: Atul Kumar	Registration Number: 12102801	
Set Assign: 02	Page Number: 01	Total Pages: 09

Question: Make a documentation or presentation on a case study-based discussion of pig. (Hint-The discussion start with the introduction, about the topic, brief history, real-time use examples, advantage, limitation, demonstration and some basic tasks)

Answer: -

INTRODUCTION



Apache Pig is a high-level data flow platform for executing MapReduce programs of Hadoop. The language used for Pig is Pig Latin. The Pig scripts get internally converted to Map Reduce jobs and get executed on data stored in HDFS. Apart from that, Pig can also execute its job in Apache Tez or Apache Spark.

Pig can handle any type of data, i.e., structured, semi-structured or unstructured and stores the corresponding results into Hadoop Data File System. Every task which can be achieved using PIG can also be achieved using java used in

MapReduce. To write data analysis programs, Pig provides a high-level language known as Pig Latin. This language provides various operators using which programmers can develop their own functions for reading, writing, and processing data. To analyse data using Apache Pig, programmers need to write scripts using Pig Latin language. All these scripts are internally converted to Map and Reduce tasks. Apache Pig has a component known as Pig Engine that accepts the Pig Latin scripts as input and converts those scripts into MapReduce jobs.

APACHE PIG- HISTORY

In 2006, Apache Pig was developed as a research project at Yahoo, especially to create and execute MapReduce jobs on every dataset. In 2007, Apache Pig was open sourced via Apache incubator. In 2008, the first release of Apache Pig came out. In 2010, Apache Pig graduated as an Apache top-level project.

WHY DO WE NEED APACHE PIG?

Programmers who are not so good at Java normally used to struggle working with Hadoop, especially while performing any MapReduce tasks. Apache Pig is a boon for all such programmers.

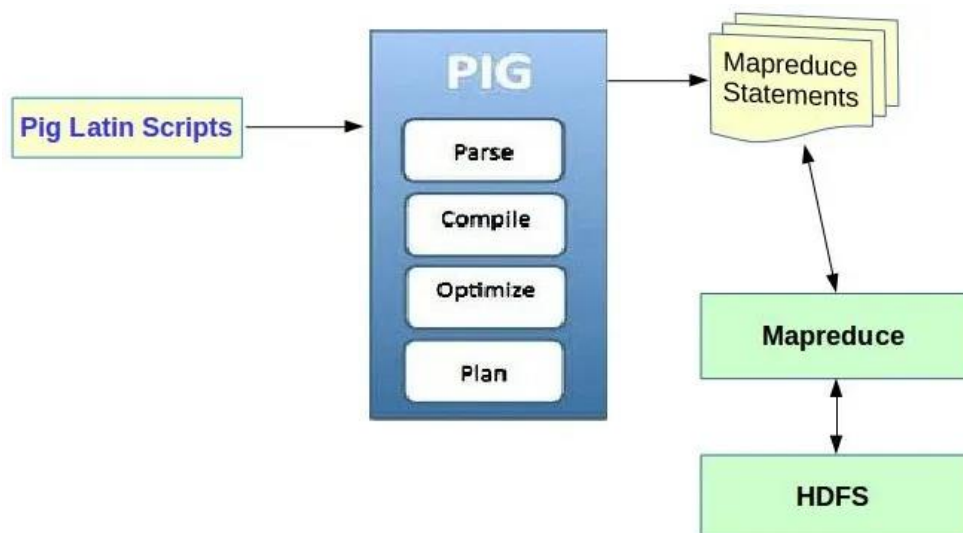
- Using Pig Latin, programmers can perform MapReduce tasks easily without having to type complex codes in Java.
- Apache Pig uses multi-query approach, thereby reducing the length of codes. For example, an operation that would require you to type 200 lines of code (LoC) in Java can be easily done by typing as less as just 10 LoC in Apache Pig. Ultimately Apache Pig reduces the development time by almost 16 times.
- Pig Latin is SQL-like language and it is easy to learn Apache Pig when you are familiar with SQL.
- Apache Pig provides many built-in operators to support data operations like joins, filters, ordering, etc. In addition, it also provides nested data types like tuples, bags, and maps that are missing from MapReduce.

ARCHITECTURE OF APACHE PIG

The language used to analyse data in Hadoop using Pig is known as **Pig Latin**. It is a high-level data processing language which provides a rich set of data types and operators to perform various operations on the data.

To perform a particular task Programmers using Pig, programmers need to write a Pig script using the Pig Latin language, and execute them using any of the execution mechanisms (Grunt Shell, UDFs, Embedded). After execution, these scripts will go through a series of transformations applied by the Pig Framework, to produce the desired output.

Internally, Apache Pig converts these scripts into a series of MapReduce jobs, and thus, it makes the programmer's job easy.



FEATURES OF APACHE PIG

➤ **Ease of Programming**

Writing complex java programs for map reduce is quite tough for non-programmers. Pig makes this process easy. In the Pig, the queries are converted to MapReduce internally.

➤ **Optimization opportunities**

It is how tasks are encoded permits the system to optimize their execution automatically, allowing the user to focus on semantics rather than efficiency.

➤ **Extensibility**

A user-defined function is written in which the user can write their logic to execute over the data set.

➤ **Flexible**

It can easily handle structured as well as unstructured data.

➤ **In-built operators**

It contains various type of operators such as sort, filter and joins.

APACHE PIG EXAMPLE

Like if we want to find the most occurred start letter using Pig.

Case 1: Load the data into bag named "lines". The entire line is stuck to element line of type character array.

```
grunt>lines = LOAD "/user/Desktop/data.txt" AS (line: chararray);
```

Case 2: The text in the bag lines needs to be tokenized this produces one word per row.

```
Grunt>tokens = FOREACH lines GENERATE flatten (TOKENIZE(line)) AS token: chararray;
```

Case 3: To retain the first letter of each word type the below command. This command uses substring method to take the first character.

```
grunt>letters= FOREACH tokens  
GENERATE SUBSTRING(0,1) as letter : chararray;
```

Case 4: Create a bag for unique character where the grouped bag will contain the same character for each occurrence of that character.

```
grunt>lettergrp = GROUP letters by letter;
```

Case 5: The number of occurrences is counted in each group.

```
grunt>countletter= FOREACH lettergrp GENERATE group , COUNT(letters);
```

Case 6: Arrange the output according to count in descending order using the commands below.

```
grunt>OrderCnt = ORDER countletter BY $1 DESC;
```

Case 7: Limit to One to give the result.

```
grunt> result =LIMIT OrderCnt 1;
```

Case 8: Store the result in HDFS . The result is saved in output directory under pig folder.

```
grunt> STORE result into 'home/pig/output';
```

ADVANTAGES OF APACHE PIG

- **Less code** - The Pig consumes less line of code to perform any operation.

- **Reusability** - The Pig code is flexible enough to reuse again.
- **Nested data types** - The Pig provides a useful concept of nested data types like tuple, bag, and map.

DISADVANTAGES OF APACHE PIG

- **Errors of pig** - Errors that Pig produces due to UDFs (Python) are not helpful at all. At times, while something goes wrong, it just gives the error such as exec error in UDF, even if the problem is related to syntax or the type error, it lets alone a logical one.
- **Delay in execution** - Unless either we dump or store an intermediate or final result the commands are not executed. This increases the iteration between debug and resolve the issue.
- **Support-** Generally, Google and Stack Overflow do not lead good solutions for the problems.

APACHE PIG - BASIC PROGRAM STRUCTURE

Execution Modes

- **Local**

- Executes in a single JVM
- Works exclusively with local file system
- Great for development, experimentation and prototyping.

- **Hadoop Mode**

- Also known as MapReduce mode.
- Pig renders Pig Latin into MapReduce jobs and executes them on the cluster.
- Execute against semi-distributed or fully-distributed Hadoop installation.

